

Experiment 7: Travelling Salesman Problem (TSP)

Aim

To solve the Travelling Salesman Problem using Python.

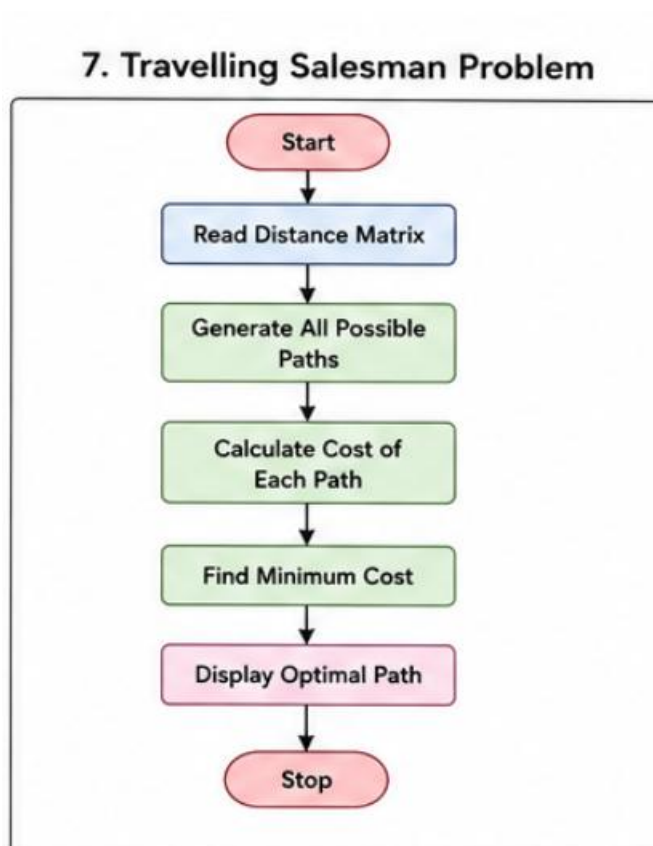
Objective

To find the minimum cost path that visits every city exactly once.

Algorithm

1. Define the distance matrix.
2. Generate all possible paths.
3. Calculate the cost of each path.
4. Find the minimum cost.
5. Display the optimal path.

Flowchart



Code

```
cities = [0,1,2]
```

```
cost = [[0,10,15],
        [10,0,20],
        [15,20,0]]
```

```
min_cost = float('inf')
```

```
for p in permutations(cities):
```

```
    total = 0
```

```
    for i in range(len(p)-1):
```

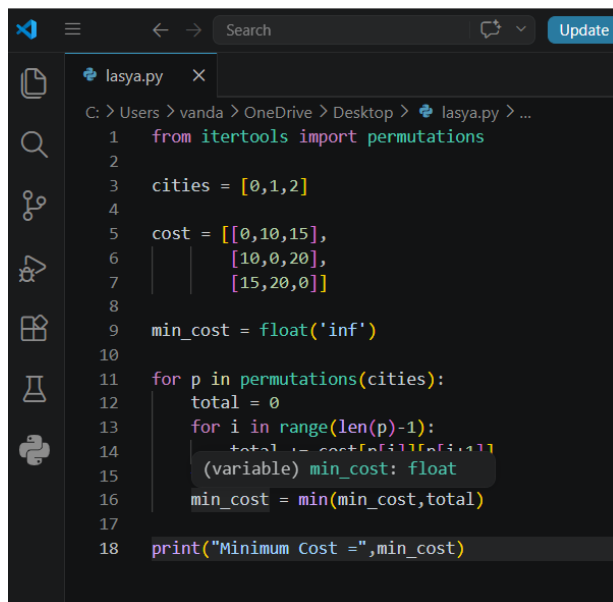
```
        total += cost[p[i]][p[i+1]]
```

```
    total += cost[p[-1]][p[0]]
```

```
    min_cost = min(min_cost,total)
```

```
print("Minimum Cost =",min_cost)
```

Output :



```
lasya.py
C:\Users\vanda> OneDrive\Desktop > lasya.py > ...
1  from itertools import permutations
2
3  cities = [0,1,2]
4
5  cost = [[0,10,15],
6         [10,0,20],
7         [15,20,0]]
8
9  min_cost = float('inf')
10
11  for p in permutations(cities):
12      total = 0
13      for i in range(len(p)-1):
14          total += cost[p[i]][p[i+1]]
15      (variable) min_cost: float
16      min_cost = min(min_cost,total)
17
18  print("Minimum Cost =",min_cost)
```

```
A B D E C F
PS C:\Users\vanda\onedrive\desktop> PYTHON lasya.py
Minimum Cost = 45
PS C:\Users\vanda\onedrive\desktop> █
```

Result

The minimum travelling cost was found successfully.

Conclusion

The program finds the shortest route covering all cities.